WATER HYDRAULIC MOTORS / PRODUCT INFORMATION

M6 Water Hydraulic Motor

Evaluate the M6 at the required starting torque and running speed. Use its displacement and performance data to confirm flow, pressure difference and drive behaviour; include case-drain limits, stopping and shaft-load requirements.

The M6 motor bridges compact and high-torque designs, delivering double the torque of the M3 while maintaining low weight at 1.2 kg (2.65 lb) for applications requiring more power without sacrificing environmental compliance.



M6 Water Hydraulic Motor

At a glance

Parameter / value	Parameter / value
Displacement 6.0 cc/rev (0.366 in³/rev)	Max Pressure 160 bar (2,321 psi) continuous
Peak Efficiency 80 %	Speed Range 1000-3000 RPM
Max Torque 14.2 Nm (10.47 lbf·ft)	Max Power 4.4 kW (5.90 hp)

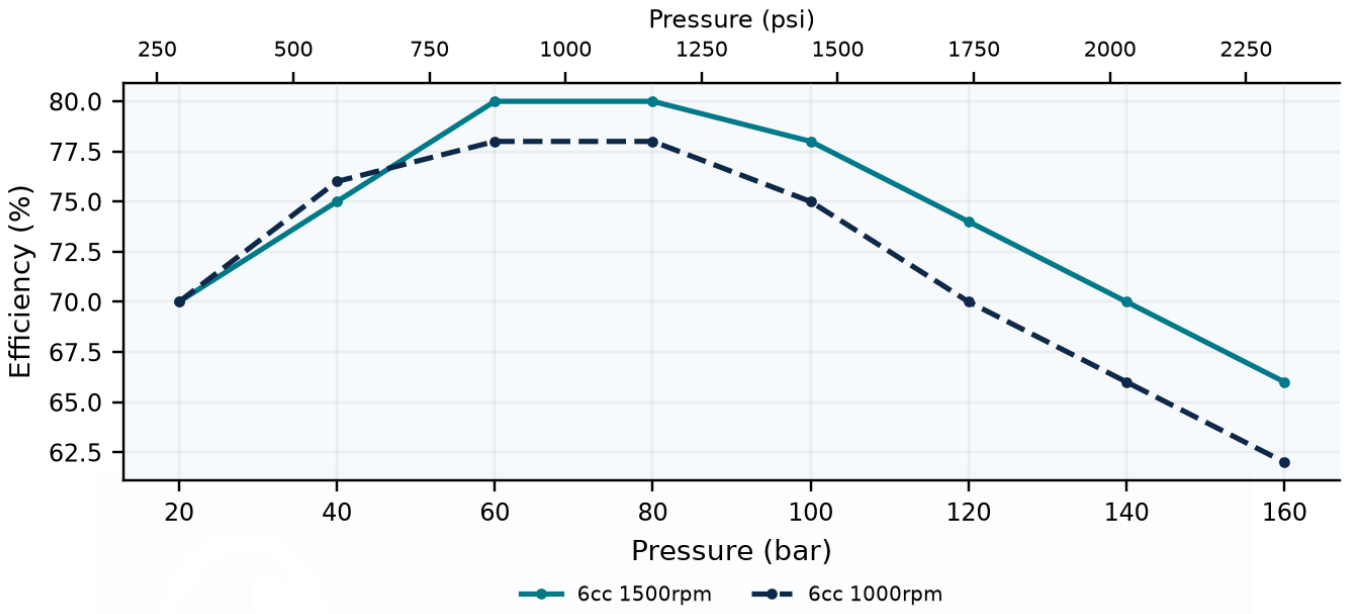
Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

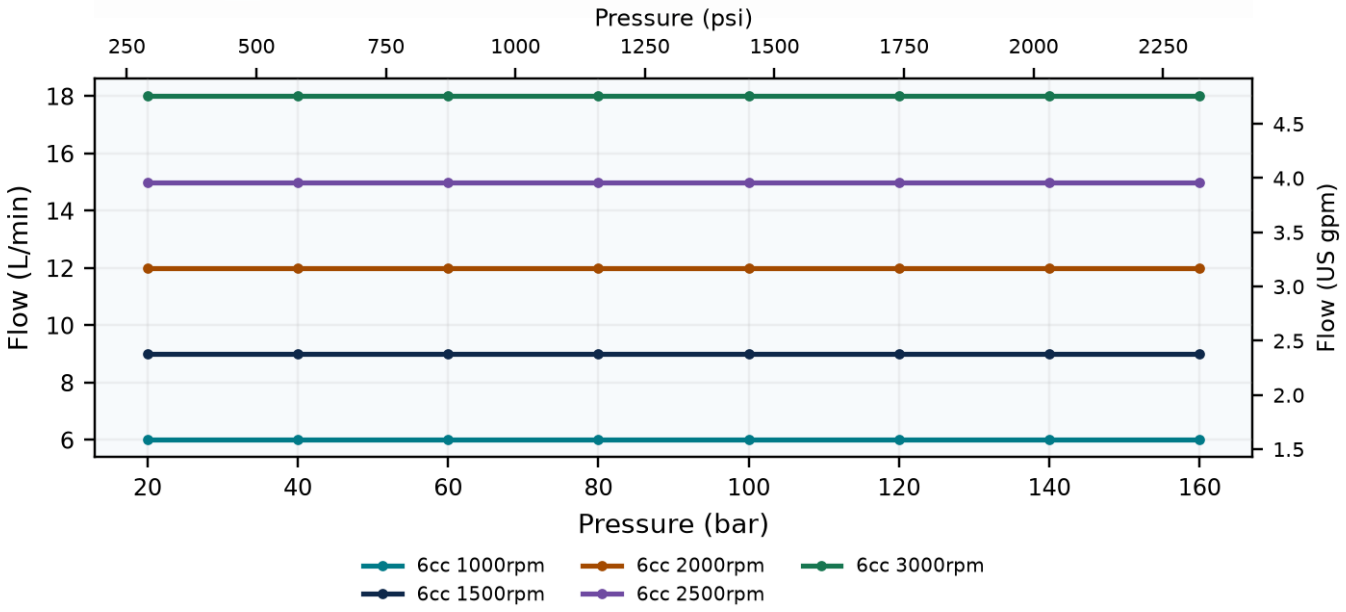
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

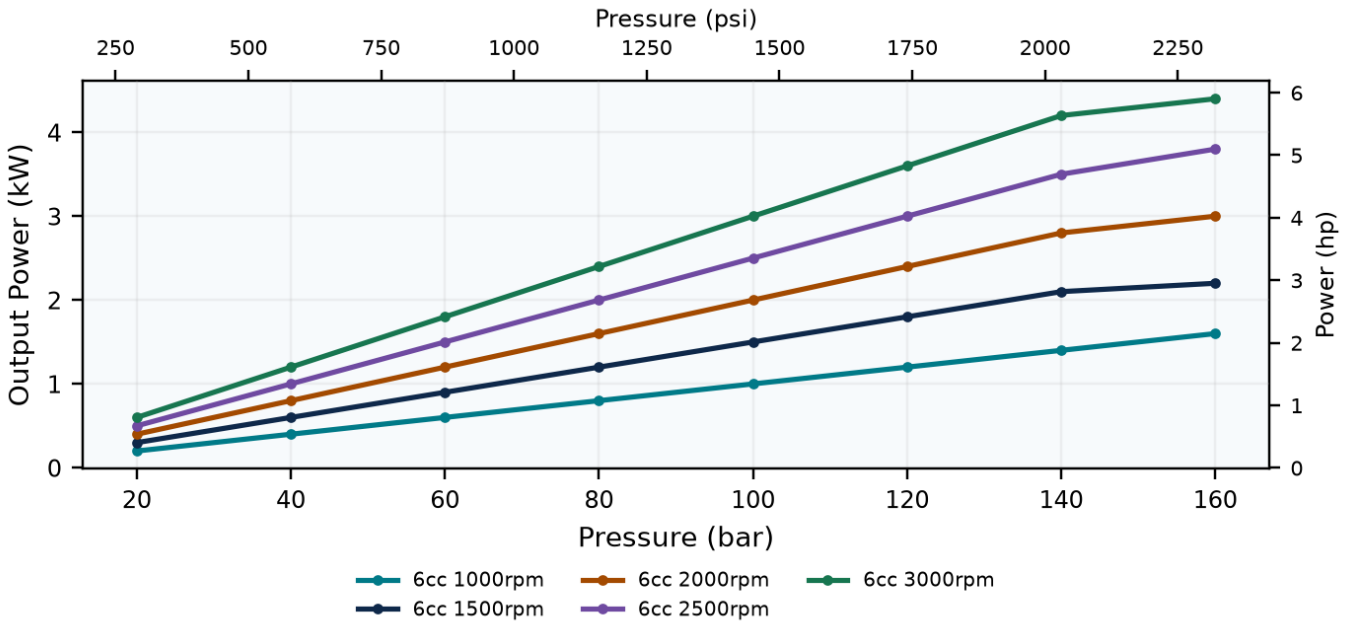
Efficiency (%) vs Pressure (bar)



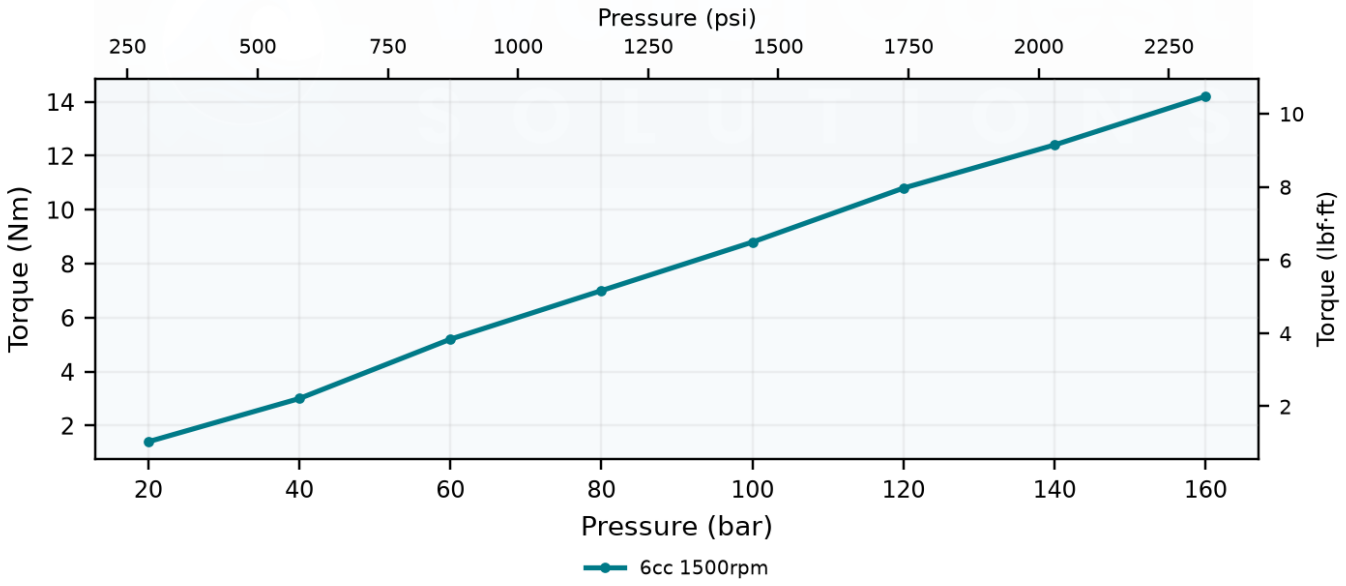
Flow (L/min) vs Pressure (bar)



Output Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)

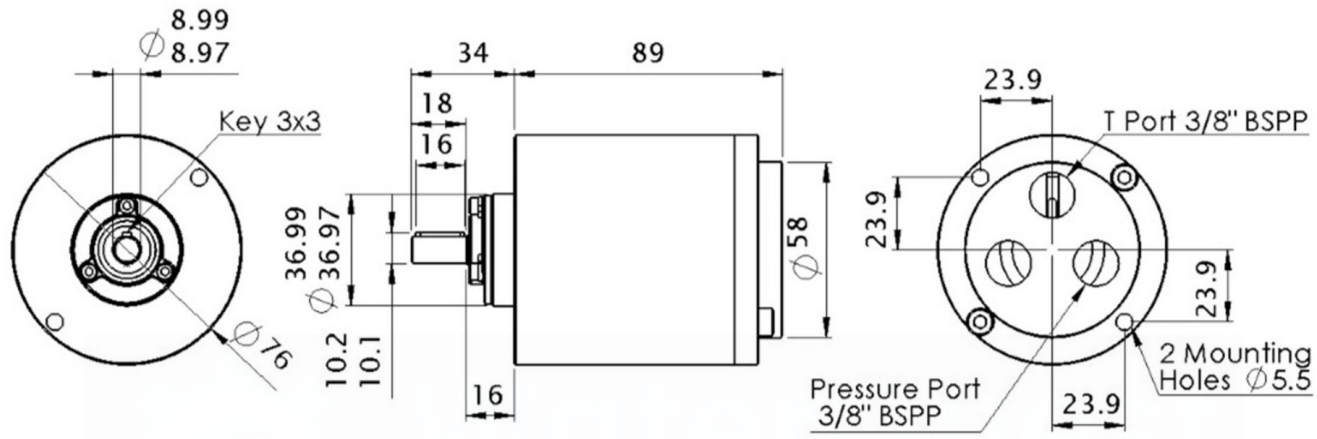


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

MB160-6.0W Technical Drawing

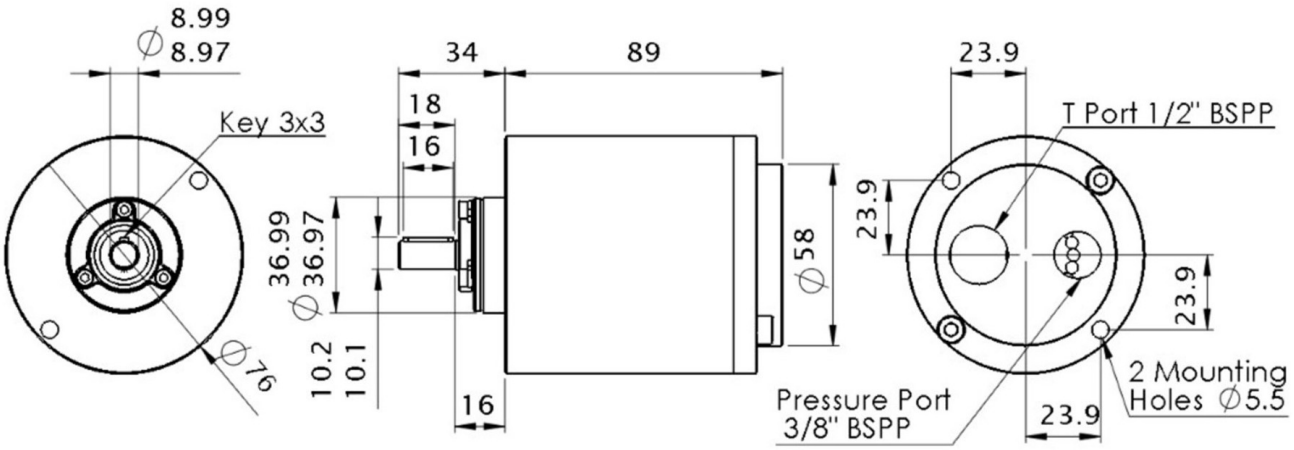
MB160-6.0W & MB160-4.6W



MC160-6.0W Technical Drawing

MC160-6.0W & MA160-6.0W

MC160-4.6W & MA160-4.6W



Application and specification

Select the wetted material and seal package for the stated chemistry, salinity, temperature and exposure. Standard stainless construction is not a universal seawater approval; confirm the exact configuration with WaterQuest.

Removing hydraulic oil from this circuit removes that oil inventory from a potential release. Water quality, additives, other machine fluids and local discharge requirements still need review for the application.

80% efficiency maintained across 40-100 bar (580-1,450 psi) range provides predictable performance for variable-load applications in processing equipment.

Specify the working-water chemistry and external exposure, then confirm the permitted material and seal package. Seawater operation requires explicit approval for the configuration; it cannot be inferred from standard stainless construction.

14.2 Nm (10.5 lbf·ft) torque capacity supports larger conveyor drives, mixers, and material handling systems in food and pharmaceutical facilities.

MB series provides reversible rotation for positioning systems and reversible conveyors without valve manifold complexity.